

1. If ω be complex cube root of unity satisfying the relation $\frac{1}{a+\omega} + \frac{1}{b+\omega} + \frac{1}{c+\omega} = 2\omega^2$ and $\frac{1}{a+\omega^2} + \frac{1}{b+\omega^2} + \frac{1}{c+\omega^2} = 2\omega$ then value of $\frac{1}{a+1} + \frac{1}{b+1} + \frac{1}{c+1}$ is equal to-
 (a) 2 (b) -2 (c) $-1 + \omega^2$ (d) $-1 + \omega$
2. If $x_r = \cos \frac{\pi}{2^r} + i \sin \frac{\pi}{2^r}$, $z_t = \cos \frac{\pi}{3^t} + i \sin \frac{\pi}{3^t}$ $r = 1, 2, 3, \dots, t = 1, 2, 3, \dots$ The value of $(x_1 x_2 x_3 \dots \infty)^2 (z_1 z_2 z_3 \dots \infty)^4$ -
 (a) 0 (b) 1 (c) -1 (d) i
3. $(z-1)(\bar{z}-1)$ can be written as
 (a) $z\bar{z} + 1$ (b) $|z|^2 + 1$ (c) $|z-1|^2$ (d) $|z|^2 + 2$
4. If $x = a + b$, $y = a\omega + b\omega^2$ and $z = a\omega^2 + b\omega$ then which one of the following is true -
 (a) $x^3 + y^3 + z^3 = 3(a^3 + b^3)$ (b) $xyz = 2(a^3 + b^3)$
 (c) $x + y + z \neq 0$ (d) $x^2 + y^2 + z^2 = a^2 + b^2$
5. If $z + \frac{1}{z} + 1 = 0$, then $z^{2003} + \frac{1}{z^{2003}}$ is equal to-
 (a) 1 (b) -1 (c) 0 (d) None of these
6. If z_1 and z_2 are two non-zero complex numbers such that $|z_1 + z_2| = |z_1| + |z_2|$, then $\text{Arg } z_1 - \text{Arg } z_2 =$
 (a) $-\pi$ (b) $-\pi/2$ (c) 0 (d) $\pi/2$
7. The equation $z^2 = \bar{z}$ has -
 (a) No solution (b) 2 solutions
 (c) Four solutions (d) An infinite number of solutions
8. If $x_n = \cos \frac{\pi}{2^n} + i \sin \frac{\pi}{2^n}$, then $\prod_{n=1}^{\infty} x_n$ is equal to -
 (a) -1 (b) 1 (c) $\frac{1}{\sqrt{2}}$ (d) $\frac{i}{\sqrt{2}}$
9. The real value of θ for which the expression $\frac{1+i\cos\theta}{1-2i\cos\theta}$ is a real number, is
 (a) $n\pi + \pi/2$ (b) $n\pi - \pi/2$
 (c) $n\pi \pm \pi/2$ (d) None of these
10. The real values of x and y for which the equation $(x^4 + 2xi) - (3x^2 + yi) = (3-5i) + (1+2yi)$ is satisfied, are
 (a) $x=2, y=3$ (b) $x=-2, y=\frac{1}{3}$
 (c) Both (a) and (b) (d) None of these
11. If $z = x + iy$, then area of the triangle whose vertices are points z , iz and $z + iz$ is
 (a) $2|z|^2$ (b) $\frac{1}{2}|z|^2$ (c) $|z|^2$ (d) $\frac{3}{2}|z|^2$
12. The inequality $|z-4| < |z-2|$ represents the region given by
 (a) $\text{Re}(z) > 0$ (b) $\text{Re}(z) < 0$
 (c) $\text{Re}(z) > 2$ (d) None of these
13. If $|z-2| = \min\{|z-1|, |z-5|\}$, where z is a complex number, then-
 (a) $\text{Re}(z) = \frac{3}{2}$ (b) $\text{Re}(z) = \frac{7}{2}$
 (c) $\text{Re}(z) \in \left\{\frac{3}{2}, \frac{7}{2}\right\}$ (d) None of these
14. For positive integers n_1, n_2 the value of the expression $(1+i)^{n_1} + (1+i^3)^{n_1} + (1+i^5)^{n_2} + (1+i^7)^{n_2}$, where $i = \sqrt{-1}$, is a real number if and only if
 (a) $n_1 = n_2 + 1$ (b) $n_1 = n_2$
 (c) $n_1 = n_2 - 1$ (d) All values of n_1 and n_2
15. If $|z^2 - 1| = |z|^2 + 1$ then z lies on a -
 (a) Circle (b) Parabola
 (c) Ellipse (d) None of these
16. In the above question the maximum value of $\angle PAB + \angle PBC + \angle PCD + \angle PDA$ is equal to-
 (a) $\frac{3\pi}{4}$ (b) $\frac{\pi}{4}$ (c) $\frac{5\pi}{4}$ (d) $\frac{\pi}{2}$
17. $Z_1 \neq Z_2$ are two points in an Argand plane. If $a|Z_1| = b|Z_2|$, then the point $\frac{aZ_1 - bZ_2}{aZ_1 + bZ_2}$ is -
 (a) In the I quadrant (b) In the III quadrant
 (c) On the real axis (d) On the imaginary axis
18. If z_1, z_2, z_3 are three complex numbers satisfying equations $|z+1| \leq 1, |z+2| \leq 2$ and $|z+4| \leq 4$ respectively then the maximum value of $|z_1| + |z_2| + |z_3|$ is
 (a) 7 (b) 10 (c) 12 (d) 14
19. If $|z-4+3i| \leq 1$ and α and β be the least and greatest values of $|z|$ and k be the least value of $\frac{x^4 + x^2 + 4}{x}$ on the interval $(0, \infty)$ then k is equal to -
 (a) α (b) β (c) $\alpha + \beta$ (d) None of these
20. For all complex numbers z_1, z_2 satisfying $|z_1| = 12$ and $|z_2 - 3 - 4i| = 5$, the minimum value of $|z_1 - z_2|$ is -
 (a) 0 (b) 2 (c) 7 (d) 17